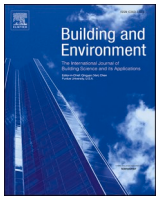
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Comparative evaluation of large language model prompting configurations for occupant-centric thermal control

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ABSTRACT

Occupant thermal demands vary with age, gender, and body mass index (BMI), yet conventional fixed-temperature and Predicted Mean Vote (PMV)-based control strategies rely on standardized parameters that do not directly accommodate these characteristics. Large language models (LLMs) can interpret occupant information and indoor environmental variables to generate thermal control setpoints, but output quality is sensitive to prompting configuration. This study systematically evaluates eight LLM-based prompting configurations for occupant-centric thermal setpoint generation. Static performance was assessed using 1080 samples from the ASHRAE Global Thermal Comfort Database II, with numerical agreement against an observed neutral-condition reference and PMV Comfort Compliance Rate (PCCR), defined as PMV-based comfort-bound compliance. The static evaluation showed that Decomposition achieved the lowest mean absolute error (0.43 °C), whereas Modular Reasoning, Knowledge, and Language (MRKL)-style modular prompting achieved the highest PCCR (91.6 %); matched cross-model analysis showed the lowest numerical errors for Decomposition across all three models, while PCCR rankings varied by model. Dynamic performance was further examined in EnergyPlus for four selected configurations: the instruction-only baseline, Decomposition, MRKL-style modular prompting, and a hybrid of the latter two. Among these configurations, the static performance ordering was not preserved under closed-loop operation. The two PMV-anchored configurations showed higher PMV-based comfort satisfaction than the other two dynamically tested configurations, and the hybrid achieved the highest aggregate comfort satisfaction among the four. These findings indicate that numerical agreement, PMV-based comfort-bound compliance, and dynamic control behavior represent distinct evaluation dimensions and characterize configuration-dependent trade-offs under the tested conditions.

1. Introduction

1.1. Background

Indoor thermal environment control aims to optimize building energy efficiency while providing occupants with a comfortable indoor environment [1,2]. While passive approaches such as envelope life-cycle assessment and orientation-based design have also contributed to improving building energy performance [3–5], active control of the indoor thermal environment remains central to achieving this balance. Within the built environment domain, two primary control strategies have been widely adopted: fixed-temperature control, which maintains a constant temperature setpoint, and control based on the Predicted Mean Vote (PMV) index, derived from a human heat balance model [6–8]. However, both strategies exhibit notable limitations. Fixed-temperature control has limited ability to respond to variations in occupant thermal demands. PMV-based control incorporates variables such as metabolic rate (MET) and clothing insulation (CLO), but it largely relies on standardized heat-balance inputs. As a result, PMV-based control provides limited direct representation of occupant characteristics, including age, gender, and body mass index (BMI), in its standard formulation [9–11].

In practice, occupant thermal demands within a space vary according to numerous factors, including age distribution, gender composition, and BMI [12–14]. Previous studies have shown that these characteristics may influence thermal perception and preferred indoor conditions, even under similar environmental conditions [15,16]. However, such information is not straightforward to incorporate into PMV-based control logic because these variables are not direct inputs in the standard

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Nomenclature

Symbols

T_a	Air Temperature [°C]
RH	Relative Humidity [%]
T_{LLM}	LLM-generated setpoint [°C]
$neutral_ta$	PMV-neutral temperature used as a comfort-bound anchor [°C]
n	Number of samples
T_r	Mean Radiant Temperature [°C]
V_a	Air Velocity [m/s]
T_{ref}	Observed neutral-condition reference temperature [°C]
i	Sample index

Abbreviations

API	Application programming interface
BMI	Body mass index [kg/m^2]
CLO	Clothing insulation [clo]
CSR	PMV-based comfort satisfaction rate [%]
GPT	Generative pre-trained transformer
HVAC	Heating, ventilation, and air conditioning
LLM	Large language model
MET	Metabolic rate [met or W/m^2]
PCCR	PMV comfort compliance rate [%]
PPD	Predicted percentage of dissatisfied [%]
ReAct	Reasoning and acting
SD	Standard deviation [°C]
TSV	Thermal sensation vote
ASHRAE	American Society of Heating, Refrigerating and Air-Conditioning Engineers
CDD	Cooling degree days
CoT	Chain of thought
DOE	U.S. Department of Energy
HDD	Heating degree days
ISO	International Organization for Standardization
MAE	Mean absolute error [°C]
MRKL	Modular reasoning, knowledge, and language
PMV	Predicted mean vote
RAG	Retrieval-Augmented Generation
RMSE	Root mean square error [°C]
ToT	Tree of thought
VAV	Variable air volume

heat-balance formulation. This limitation arises not solely from a lack of data but also from the structural rigidity of conventional equation-based control frameworks. Variations in thermal preference associated with age, gender, and BMI are related to complex physiological mechanisms, including differences in metabolic characteristics, thermoregulation, and body heat exchange. These mechanisms can involve nonlinear interactions among occupant and indoor environmental variables, making them difficult to represent through a single standardized equation. Consequently, uniform indoor temperature control that fails to account for occupant-related diversity may compromise thermal comfort for some occupants and contribute to unnecessary heating and cooling loads. Addressing these inefficiencies has motivated data-driven efforts to improve building energy and carbon performance, including machine learning-based heating energy forecasting, carbon-aware optimization frameworks, and life-cycle carbon modeling of building components [17–19].

To address these limitations, recent studies have explored the integration of the reasoning capabilities of large language models (LLMs) into building operation and control logic [20–22]. LLMs can process heterogeneous inputs, follow structured instructions, and infer context-dependent relationships among variables without requiring all relationships to be explicitly predefined in equation form [23]. In thermal control, this capability can be used as a reasoning interface that jointly interprets occupant information and indoor environmental variables to generate thermal control setpoints. However, LLM outputs vary considerably depending on input prompt structure, contextual information, and reasoning format [24,25]. Therefore, systematic evaluation is needed to clarify how different prompting configurations affect numerical agreement, PMV-based comfort-bound compliance, and dynamic control behavior in LLM-based occupant-centric thermal control.

1.2. Literature review

Within the built environment domain, LLM reasoning capabilities have been applied across diverse operational contexts, including building metadata interpretation, fault detection and diagnostics, occupancy measurement, and indoor environmental informatics [26–29]. Prior studies have highlighted the applicability of LLMs in domains involving heterogeneous or unstructured data that are difficult to address with conventional numerical algorithms. Examples include the automation of building performance analysis through the integration of LLMs with multi-agent systems [30] and the identification of indoor environmental quality requirements through the analysis of unstructured occupant feedback text [31].

This line of research has also extended to indoor thermal environments and occupant comfort control. Representative applications include using LLMs to infer passenger thermal sensations in large-scale spaces, such as airport terminals, for heating, ventilation, and air-conditioning (HVAC) setpoint optimization [32], integrating biometric data with LLMs to predict thermal comfort [33], and designing indoor air distribution systems with expert-level alternatives subsequently validated via computational fluid dynamics [34]. Collectively, these studies indicate that LLMs can capture complex relationships between occupant-related information and physical environmental variables. However, existing research has largely focused on prediction, feedback interpretation, or design generation, with limited attention to how LLM-generated thermal setpoints behave under different prompting configurations and dynamically changing building environments.

In parallel, research on prompt engineering — which optimizes input instructions delivered to LLMs to improve response quality and reasoning consistency — remains limited in building control applications. Because LLMs can produce different control outputs depending on prompting configuration and contextual design, even when identical data are provided, prompt design is a critical determinant of the reliability of LLM-based control systems. Although prior studies have evaluated individual prompting techniques, such as Zero-shot prompting or Chain of Thought (CoT), these investigations have typically been conducted in isolation [35]. To date, little systematic research has been conducted on how different prompting configurations affect thermal control setpoints when occupant information and indoor environmental variables are jointly provided. In particular, the relationships among numerical agreement with reference conditions, PMV-based comfort--bound compliance, and closed-loop dynamic behavior remain insufficiently understood. Therefore, a systematic comparison of representative prompting configurations is required to clarify their strengths, limitations, and applicability to LLM-based occupant-centric thermal control.

1.3. Classification of prompting techniques

Prompt engineering refers to the systematic design and optimization of input instructions to elicit appropriate outputs from artificial intelligence models [36]. While early approaches were limited to providing simple commands or illustrative examples, prompting techniques have since been developed to leverage LLM reasoning capabilities, with more than 50 distinct techniques reported in the literature [37]. These techniques can be broadly classified into six categories based on their objectives and operational mechanisms (Table 1), each exhibiting distinct strengths and limitations [38,39].

The first category, Instruction/Example, represents the most fundamental form, in which the model is directly provided with the task and reference examples. This category includes Zero-shot prompting, which issues instructions without any examples, and One-shot and Few-shot prompting techniques, which provide one or more examples. The second category, Reasoning, guides the model to solve complex problems sequentially along a structured reasoning path. Representative techniques include CoT and Tree of Thought (ToT), which are particularly valuable for tasks requiring explicit logical organization. The third category, Context/Knowledge, augments the model response with external knowledge, fixed contextual information, or tool-like modular structures. Techniques in this category, such as Retrieval-Augmented Generation (RAG), Reasoning and Acting (ReAct), and Modular Reasoning, Knowledge, and Language (MRKL), are commonly discussed as ways to improve grounding or task decomposition. Beyond these, three additional categories have been identified: the Optimization category, which automatically searches for and refines prompt quality; the Adaptive/Role category, which adjusts tone and perspective to suit specific roles or contexts; and the Meta-Prompting category, in which the model itself generates or evaluates prompts.

In complex control contexts such as indoor thermal environment regulation, effective setpoint generation requires the simultaneous interpretation of occupant information and physical environmental variables. Prompting configurations that combine structured reasoning and contextual knowledge are directly relevant to examining how LLMs generate thermal control setpoints from mixed inputs. Specifically, the Reasoning category can structure the model's interpretation process, while the Context/Knowledge category can incorporate domain knowledge or modular comfort-bound references. This study therefore selects representative prompting configurations from these two categories for quantitative comparative analysis, clarifying how different prompting configurations lead to distinct characteristics in numerical agreement, PMV-based comfort-bound compliance, and dynamic control behavior. The primary classification frameworks and representative techniques within each category are summarized in Table 1.

Table 1
Classification and characteristics of prompt engineering categories.

Category	Representative Techniques	Characteristic & Strengths	Limitations
Instruction/ Example	Zero-shot, Few-shot	Intuitive and easy to implement, leveraging baseline model behavior	Potential example bias and limited effectiveness in complex logical tasks
Reasoning	CoT, ToT, Decomposition	Supports logical inference through step-by-step or multi-path reasoning	Increased response latency and potential inconsistencies in reasoning
Context/ Knowledge	RAG, ReAct, MRKL	Supports knowledge-grounded or modular reasoning by incorporating contextual information or tool-like structures	Performance depends on context quality, retrieval reliability, or module design
Optimization	Active, Ensembling	Improves prompt structures through automated search or feedback	High computational costs and time-consuming search processes
Adaptive/ Role	Role/Persona, Dynamic	Strengthens domain-specific perspectives through role or situational framing	Risk of overfitting to specific roles, reducing generalizability
Meta-Prompting	Self-Critique, Reflection	Enhances output reliability through self-verification or prompt refinement	Slower processing speeds due to iterative evaluation steps

1.4. Research purpose

This study evaluates how LLM prompting configurations generate occupant-conditioned thermal setpoints from age, gender, BMI, and indoor environmental variables. All eight configurations were assessed statically using the ASHRAE Global Thermal Comfort Database II based on reference-temperature agreement, PMV Comfort Compliance Rate, statistical uncertainty, cross-model sensitivity, PMV-filter sensitivity, and a controlled counterfactual analysis of occupant metadata. Four selected configurations—B1, R3, C3, and R3+C3—were then examined in closed-loop EnergyPlus simulations using PMV-based comfort satisfaction, setpoint variability, and HVAC electricity use. The U.S. Department of Energy (DOE) prototype thermostat schedule was included only as a schedule-based contextual reference. This framework distinguishes the evidence obtained from the eight-configuration static comparison from that obtained from the four-configuration dynamic evaluation.

2. Prompting configurations

In this study, the term "prompting configuration" refers to a standardized prompt structure designed to generate thermal control setpoints from occupant information and indoor environmental variables under a common input-output format. Reasoning- and Context/Knowledge-category configurations were selected because setpoint generation requires both structured interpretation of occupant information and consistency with thermal comfort knowledge. The former organizes intermediate reasoning steps over occupants' physiological and environmental variables, while the latter provides domain context or modular structures to support comfort-related decisions. This study focuses on these two categories and an Instruction-only baseline configuration; the Optimization, Adaptive/Role, and Meta-Prompting categories are left for future investigation, as these categories primarily address prompt search, persona framing, or self-evaluation. The design intent and structural characteristics of the seven initial configurations are summarized in Table 2.

(1) Baseline configuration (B1, Instruction-only): Established as a reference case to assess LLM-generated setpoints when only input variables and the required output format are provided, without guidance along a reasoning path or augmentation with external contextual information.
(2) Reasoning-category configurations (R1-R3):

- Zero-shot CoT (R1): Guides the model to produce step-by-step reasoning before generating the final setpoint and evaluates whether sequential reasoning improves the interpretation of occupant information and environmental variables.
- Tree of Thought (R2): Structures the response through multiple expert-like reasoning paths and then integrates the proposed setpoints. This configuration is used to examine decision-making stability under complex conditions involving potentially conflicting thermal factors.
- Decomposition (R3): Solves the overall task by dividing it into Occupant Analysis, Environmental Analysis, and Synthesis stages. This configuration examines whether sub-task separation improves

Table 2
Initial prompting configurations selected for comparative evaluation.

Category	ID	Configuration	Objective in Thermal Setpoint Generation	Expected Structural Benefit
Baseline	B1	Instruction-only	To establish a reference case without explicit reasoning guidance	Direct setpoint generation from input variables
Reasoning	R1	Zero-shot CoT	To guide sequential reasoning over occupant and environmental variables	Step-by-step interpretation of setpoint-relevant factors
Reasoning	R2	Tree of Thought (ToT)	To generate and integrate multiple reasoning paths	Multi-perspective evaluation under conflicting thermal factors
Reasoning	R3	Decomposition	To divide the task into occupant analysis, environmental analysis, and synthesis	Structured sub-task reasoning for setpoint generation
Context/ Knowledge	C1	RAG-style prompting	To provide fixed thermal comfort knowledge within the prompt	Knowledge-grounded setpoint reasoning under controlled context
Context/ Knowledge	C2	ReAct-style prompting	To structure reasoning through Thought-Action-Observation-Decision steps	Iterative verification of setpoint decisions
Context/ Knowledge	C3	MRKL-style modular prompting	To separate PMV-neutral anchoring, occupant-trait adjustment, and final integration	Modular integration of comfort-bound anchoring and occupant-trait adjustment

numerical agreement with the reference condition while making the intermediate reasoning steps more explicit.

(3) Context/Knowledge-category configurations (C1-C3):

- RAG-style prompting (C1): Provides fixed thermal comfort knowledge extracted from ASHRAE Standard 55 and ISO 8996 [40,41] within the prompt. This configuration examines how domain knowledge affects setpoint generation under a controlled context condition.
- ReAct-style prompting (C2): Organizes the reasoning process into Thought-Action-Observation-Decision steps. In this configuration, the Action and Observation stages are represented within the prompt

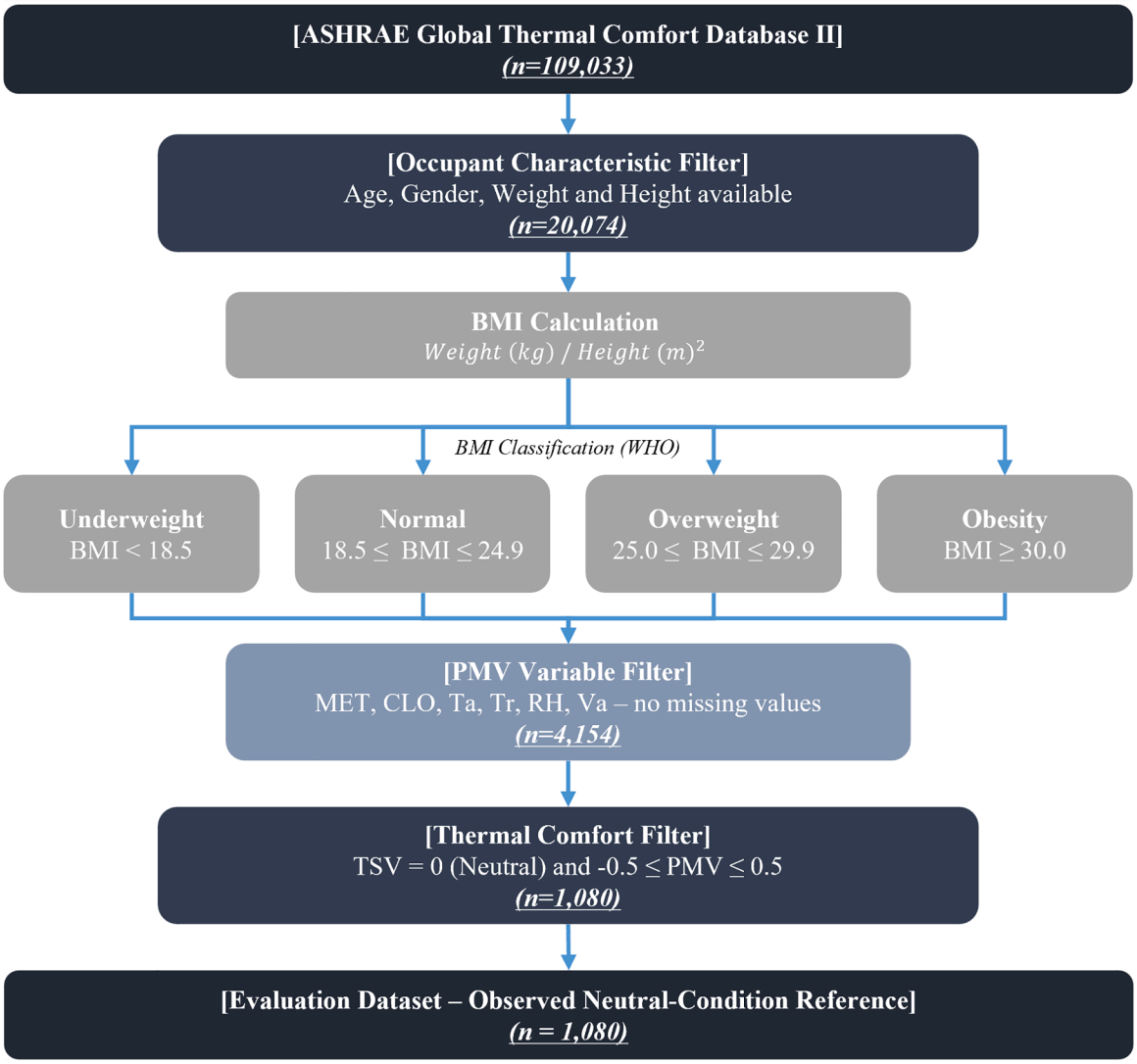


Fig. 1. Filtering process for construction of the main evaluation dataset.

structure to support iterative verification of the generated setpoint under a standardized input-output format.
- MRKL-style modular prompting (C3): Separates the setpoint-generation process into a PMV-neutral anchor, an occupant-trait adjustment module, and final integration. This configuration examines how modular separation between comfort-bound anchoring and occupant-trait adjustment affects generated setpoints.

3. Static performance evaluation

3.1. Dataset and evaluation metrics

The ASHRAE Global Thermal Comfort Database II was filtered to retain records containing age, gender, weight, height, MET, CLO, T_a, T_r, RH, V_a, thermal sensation vote (TSV), and valid PMV values. BMI was calculated as weight/height^2 and classified as Underweight (<18.5), Normal (18.5–24.9), Overweight (25.0–29.9), or Obesity (≥30.0) according to World Health Organization criteria [42]. Among TSV-neutral records, 1080 also satisfied $-0.5 \leq \text{PMV} \leq 0.5$ and formed the main neutral-condition reference dataset. The remaining 762 TSV-neutral records fell outside this PMV range and were retained as a PMV-filter sensitivity subset. Fig. 1 summarizes the construction of the main evaluation dataset.

Three primary evaluation metrics, two complementary PMV-based comfort indicators, and three implementation-related indicators—mean response time, token use, and estimated API cost—were used to characterize static setpoint-generation performance. Mean absolute error (MAE) and root mean square error (RMSE) measure numerical agreement between the LLM-generated control temperature and the observed neutral-condition reference temperature, as formally defined in Eqs. (1) and (2), respectively. PMV Comfort Compliance Rate (PCCR) quantifies the proportion of LLM-generated temperatures that satisfy the PMV comfort range ($-0.5 \leq \text{PMV} \leq 0.5$) when applied to the corresponding occupant and environmental conditions, as defined in Eq. (3). PCCR is therefore interpreted as a PMV-based comfort-bound compliance indicator rather than as prediction accuracy against an independently observed personalized-comfort label.

Mean absolute PMV and mean predicted percentage of dissatisfied (PPD) were additionally calculated at the generated setpoint to compare PCCR with established PMV-derived thermal-comfort measures. Lower mean absolute PMV indicates closer proximity to PMV neutrality, while lower mean PPD indicates a lower predicted proportion of dissatisfied occupants. Because the static dataset consists of independent observations without temporal continuity or duration, occupied comfort hours and discomfort degree-hours were not calculated for the static evaluation. Time-based comfort-bound compliance was evaluated separately using the PMV-based comfort satisfaction rate (CSR) in the dynamic simulation. Mean response time refers to the average end-to-end time required for each configuration to process one query (s/query) under the common test conditions. In Eqs. (1)–(3), n denotes the total number of evaluation samples, i indexes individual samples, $T_{LLM,i}$ is the LLM-generated setpoint for sample i, and $T_{ref,i}$ is the corresponding observed neutral-condition reference temperature. In Eq. (3), $PMV(T_{LLM,i})$ denotes the PMV value obtained by substituting $T_{LLM,i}$ for air temperature while holding the sample's other PMV inputs (T_r, RH, V_a, MET, CLO) at their observed values, and $1[\cdot]$ equals 1 when the specifiedwhen the specifiedwhen the specifiedwhen the specifiedwhen the specifiedPMV condition is satisfied and 0 otherwise.

$$MAE = \frac{1}{n}\sum_{i=1}^{n}\left|T_{LLM,i} - T_{ref,\,i}\right| \tag{1}$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(T_{LLM,i} - T_{ref,\,i}\right)^2} \tag{2}$$

$$PCCR = \frac{1}{n}\sum_{i=1}^{n} 1\left[-0.5 \leq PMV\left(T_{LLM,i}\right) \leq 0.5\right] \times 100 \tag{3}$$

3.2. Experimental setup

To ensure a fair performance comparison across the prompting configurations, the following experimental control strategy was implemented. All configurations were provided with identical input comprising four indoor environmental variables (T_a, T_r, RH, V_a) and five occupant characteristic variables (MET, CLO, Age, Gender, BMI). Furthermore, the output format was standardized as "Final result: [Value.1] °C" to eliminate extraneous variables beyond differences in prompt structure.

For the Context/Knowledge-category configurations (C1, C2, C3), standardized context and modular structures were used to maintain comparable input-output conditions across configurations. For C1 (RAG-style prompting), key knowledge from ASHRAE Standard 55 and ISO 8996 was pre-extracted into discrete knowledge chunks and directly included in the prompt. For C2 (ReAct-style prompting), the response was structured through Thought-Action-Observation-Decision steps to examine the effect of iterative verification within the prompt. For C3 (MRKL-style modular prompting), the neutral temperature at which PMV = 0 was pre-calculated using a Python-based PMV computation module (pythermalcomfort library [43]) based on the Fanger PMV model and provided as a PMV-neutral comfort-bound anchor. The subsequent reasoning process was separated into occupant-trait adjustment and final integration.

The main static evaluation was performed using the GPT-4o-2024–08–06 model via OpenAI's application programming interface (API) [44]. The temperature parameter was set to 0 to reduce decoding randomness and maintain consistent generation conditions across configurations. Each sample–configuration pair was generated once. Accordingly, the bootstrap confidence intervals described below characterize sampling variability across evaluation records conditional on the observed outputs and do not quantify run-to-run variability in LLM responses. All evaluations were conducted on a system equipped with a 12th Gen Intel Core i7–12700 K (3.61 GHz) processor, 32.0 GB RAM, running Windows 11 Pro, with Python 3.9. Nonparametric bootstrap 95 % confidence intervals for MAE, RMSE, and PCCR were calculated using 20,000 resamples. Overall differences in paired absolute and squared errors were assessed using Friedman tests with Kendall's W, followed by all 28 pairwise Wilcoxon signed-rank tests with Holm correction. Binary PCCR outcomes were assessed using Cochran's Q test followed by exact McNemar tests with Holm correction. All main analyses used 1080 complete matched records.

To provide a matched cross-model analysis, all eight configurations were compared across GPT-4o, Claude Opus 4.5, and Llama-3.3–70B-Versatile using the same 200-sample subset randomly selected from the 1080-sample dataset with a fixed random seed [45,46]. The corresponding GPT-4o outputs were extracted from the 1080-sample results using the original source indices. This analysis examined whether the overall rankings and configuration-dependent trends were reproduced across model families rather than directly ranking the models. The same omnibus and Holm-corrected post hoc framework was applied separately within each model, with all descriptive and inferential analyses based on the 200 common samples.

To examine sensitivity to the PMV-based filtering criterion, all eight prompting configurations were additionally evaluated using the 762 TSV-neutral samples whose observed PMV values fell outside the −0.5 to 0.5 range. The same model, prompting configurations, decoding settings, input–output format, and evaluation procedures were applied. Results were reported separately for the PMV-consistent main subset (n = 1080) and the PMV-inconsistent sensitivity subset (n = 762). For both subsets, MAE, RMSE, PCCR, mean absolute PMV, and mean PPD were calculated by substituting the LLM-generated setpoint for air

temperature while holding T_r, RH, V_a, MET, and CLO at their corresponding observed values. This analysis was used to determine whether the relative configuration rankings and the apparent advantage of PMV-anchored configurations depended on the PMV-based sample-selection criterion.

To isolate the contribution of occupant characteristics, a controlled counterfactual sensitivity analysis was additionally conducted using GPT-4o across all eight prompting configurations. Twelve observed conditions at MET = 1.2 were selected from the 1080-sample dataset using a 3 × 2 × 2 stratification of air temperature, CLO, and air velocity, while retaining the corresponding T_r and RH values. Within each condition, T_a, T_r, RH, V_a, MET, and CLO were held constant. Age was varied among 20, 50, and 80 years, gender between male and female, and BMI among Underweight, Normal, Overweight, and Obesity, with one characteristic varied at a time around a common reference profile of 50 years, male, and Normal BMI. After duplicate reference profiles were counted once, the seven unique profiles per condition yielded 84 counterfactual inputs per configuration (12 conditions × 7 profiles). Median setpoint differences and directional consistency were evaluated for each occupant characteristic, together with ordered response patterns across the four BMI categories. Exploratory Friedman tests were applied to age and BMI, and paired Wilcoxon signed-rank tests to gender, with Holm correction across the 24 configuration–factor comparisons. The detailed prompt templates for each configuration are provided in Table A.1 to ensure reproducibility.

3.3. Quantitative comparison results

3.3.1. Static performance and sensitivity analyses

The initial comparison of seven prompting configurations revealed notable differences between configurations that closely tracked the observed neutral-condition reference and those that achieved higher PMV-based comfort-bound compliance. Decomposition (R3, MAE = 0.43 °C) recorded the lowest observed numerical error, while MRKL-style modular prompting (C3, PCCR = 91.6 %) recorded the highest observed PMV-based comfort-bound compliance. Based on these complementary performance profiles, R3+C3 was subsequently constructed and evaluated as an eighth configuration by integrating decomposition-based reasoning with PMV-neutral anchoring. The results of the final eight-configuration comparison are presented in Table 3a.

In terms of numerical agreement, R3 recorded the lowest observed MAE (0.43 °C) and RMSE (0.60 °C), while C3 recorded the highest observed PCCR (91.6 %), followed by C2 (89.0 %). R3 yielded a comparatively low PCCR of 72.4 %, whereas C3 had a substantially higher MAE of 1.52 °C than R3. Bootstrap confidence intervals were non-overlapping for the R3–C3 contrast: the MAE confidence interval of R3 (0.41–0.46 °C) did not overlap with that of C3 (1.45–1.60 °C), and the PCCR confidence interval of C3 (89.9–93.1 %) did not overlap with that of R3 (69.7–75.1 %).

Overall differences among the eight configurations were significant for both absolute and squared errors (Friedman $\chi^2(7) = 1557.11$, $p < .001$, Kendall's W = 0.207 for each) and for PMV-based comfort-bound compliance (Cochran's $Q(7) = 392.24$, $p < .001$). Holm-corrected post hoc comparisons showed that R3 had significantly lower absolute and squared errors than most other configurations. However, the R3–C1 differences were not significant for either absolute error or squared error (adjusted p = .756 and .368, respectively). C3 recorded the highest observed PCCR, but its difference from C2 was not significant after Holm correction (adjusted p = .141). The R3–C3 differences remained significant for absolute error, squared error, and PMV-based comfort-bound compliance (all adjusted $p < .001$). These results support the distinction between numerical agreement and PMV-based comfort-bound compliance as separate evaluation dimensions.

Tree of Thought (R2), which generates multiple reasoning paths, recorded an MAE of 1.03 °C. Its MAE was lower than that of the Baseline (B1, 1.35 °C) but higher than that of the single-path Zero-shot CoT configuration (R1, 0.67 °C), indicating an intermediate level of numerical agreement. These results suggest that in convergent tasks requiring the identification of a single thermal control setpoint, the inherently divergent reasoning structure of R2 may induce inter-path averaging or compromise, thereby reducing the consistency of final outputs.

The hybrid configuration (R3+C3), designed by integrating R3 and C3, recorded an MAE of 1.50 °C, RMSE of 1.97 °C, and PCCR of 88.1 %. The hybrid configuration maintained relatively high PCCR but showed lower numerical agreement than R3 and lower PCCR than C3 (all Holm-adjusted $p < .001$). This result indicates that combining two configurations with complementary strengths can still produce a distinct performance profile shaped by the interaction between reference-temperature decomposition and PMV-neutral anchoring. A possible explanation for this performance pattern is further examined through the qualitative analysis in Section 3.4.

Table 3b presents a matched comparison of all eight configurations across GPT-4o, Claude, and Llama using the same 200 samples. R3 recorded the lowest observed MAE and RMSE in all three models, whereas the rankings of the remaining configurations, particularly for PCCR, varied across models. Global differences among the eight configurations were significant within all three models for numerical error and PMV-based comfort-bound compliance (all omnibus $p < .001$). Holm-corrected R3–C3 comparisons also remained significant across all models (all adjusted $p < .001$). These results indicate that R3's lowest numerical-error ranking was reproduced across the tested models, while the complete configuration rankings remained partly model dependent.

In the main 1080-sample subset, C3 recorded the lowest mean absolute PMV and mean PPD (0.16 and 6.39 %, respectively), while R3+C3 showed lower values than C2 (0.19 and 6.97 % vs. 0.28 and 7.76 %) despite a slightly lower PCCR (88.1 %vs. 89.0 %), indicating that threshold compliance and average proximity to PMV neutrality provided related but not identical information. A sensitivity analysis was conducted using the 762 TSV-neutral samples whose observed PMV values fell outside the −0.5 to 0.5 range. The leading configuration under each primary evaluation dimension remained unchanged. R3 retained the lowest numerical error (MAE = 0.47 °C; RMSE = 0.59 °C),

Table 3a
Static performance of the eight prompting configurations.

Category	ID	Configuration	Mean setpoint (°C)	MAE (°C)[a]	RMSE (°C)[a]	PCCR (%)[a]
Baseline	B1	Instruction-only	24.82	1.35 (1.26–1.45)	2.06 (1.94–2.18)	81.9 (79.6–84.2)
Reasoning	R1	Zero-shot CoT	25.56	0.67 (0.63–0.72)	1.00 (0.90–1.12)	85.7 (83.6–87.8)
	R2	Tree of Thought (ToT)	25.22	1.03 (0.96–1.09)	1.51 (1.40–1.61)	86.0 (83.9–88.1)
	R3	Decomposition	26.18	0.43 (0.41–0.46)	0.60 (0.53–0.70)	72.4 (69.7–75.1)
Context/Knowledge	C1	RAG-style	25.91	0.62 (0.56–0.68)	1.17 (1.06–1.28)	76.0 (73.4–78.6)
	C2	ReAct-style	25.48	0.68 (0.62–0.74)	1.19 (1.10–1.28)	89.0 (87.0–90.8)
	C3	MRKL-style modular	25.36	1.52 (1.45–1.60)	1.99 (1.85–2.13)	91.6 (89.9–93.1)
Hybrid	R3+C3	Decomposition + MRKL-style modular	25.47	1.50 (1.42–1.57)	1.97 (1.84–2.10)	88.1 (86.1–90.0)

[a] Values in parentheses indicate 95 % bootstrap confidence intervals. Descriptive metrics and inferential comparisons were based on all 1080 samples for each configuration.

Table 3b
Matched cross-model sensitivity analysis of all eight prompting configurations using 200 common samples.

Configuration	Model	MAE (°C)	RMSE (°C)	PCCR (%)
B1	Llama	1.54	2.22	78.5
	Claude	1.86	2.29	78.5
	GPT-4o	1.40	2.06	80.5
R1	Llama	1.44	1.94	81.5
	Claude	1.26	1.62	79.0
	GPT-4o	0.73	1.23	85.0
R2	Llama	1.91	2.49	72.0
	Claude	1.43	1.93	84.5
	GPT-4o	1.07	1.60	84.0
R3	Llama	0.77	1.33	75.5
	Claude	0.88	1.25	76.5
	GPT-4o	0.44	0.58	75.5
C1	Llama	1.55	2.25	79.5
	Claude	1.37	1.80	92.0
	GPT-4o	0.63	1.20	72.0
C2	Llama	1.32	2.14	82.0
	Claude	1.37	1.80	89.0
	GPT-4o	0.71	1.23	87.0
C3	Llama	1.70	2.12	90.5
	Claude	1.69	2.15	91.0
	GPT-4o	1.57	1.92	91.5
R3+C3	Llama	1.65	2.01	88.0
	Claude	1.60	2.03	88.5
	GPT-4o	1.56	1.90	88.0

Note: All descriptive and inferential comparisons were based on the same 200 common samples for each model.

whereas C3 retained the highest PCCR (89.4 %), followed by R3+C3 (88.3 %). However, C3's MAE increased from 1.52 °C in the PMV-consistent subset to 4.66 °C in the PMV-inconsistent subset, while R3 retained low numerical error but recorded a PCCR of 14.8 %. The highest PCCR among the configurations without a PMV-neutral anchor was 31.8 % for B1. The continuous PMV-derived indicators showed the same general pattern. In the PMV-inconsistent subset, C3 recorded a mean absolute PMV of 0.19 and a mean PPD of 6.89 %, followed by R3+C3 at 0.21 and 7.23 %, respectively, whereas R3 recorded 0.99 and 28.78 %. Thus, inclusion of the PMV-inconsistent TSV-neutral samples did not reverse the metric-specific leading rankings, but showed that PMV-based compliance measures favored configurations incorporating a PMV-neutral anchor and differed from agreement with observed TSV-neutral temperatures.

The controlled counterfactual analysis showed that changes in age, gender, and BMI altered the generated setpoints in several configurations, although the magnitude and directional consistency of these responses varied substantially across prompting configurations (Table B.1). C3 showed the clearest response: increasing age from 20 to 80 years increased the median setpoint by 0.75 °C, and changing gender from male to female increased it by 0.60 °C, with the same direction observed under all 12 fixed environmental conditions. Relative to Normal BMI, C3 generated median setpoint changes of +0.50 °C for Underweight, −0.15 °C for Overweight, and −0.40 °C for Obesity. R3+C3 produced a fully consistent age response and median BMI shifts in the prompt-defined direction, while R1 showed comparatively stable age- and BMI-conditioned changes. After Holm correction, differences remained statistically distinguishable for all three characteristics in C3, for age and BMI in R1 and R3+C3, and for BMI in R2. By contrast, C2 and R3 were largely insensitive or directionally inconsistent, and R2 produced BMI-related changes that did not follow a coherent expected direction. These findings indicate that occupant characteristics influenced setpoint generation in several configurations, but the magnitude and consistency of these responses depended strongly on prompting configuration.

3.3.2. Response time, token use, and API cost

Table 4 summarizes mean response time, token use, and estimated API cost per query under the common test conditions. R3 and C2 showed the longest mean response times (11.32 and 11.26 s/query, respectively), while R3 also produced the most output tokens (549) and had the highest estimated cost (USD 0.0059/query). C3 had the lowest estimated cost (USD 0.0028/query) and the fewest output tokens (241), whereas R3+C3 used the most input tokens (295) but showed the shortest mean response time (5.12 s/query). These results indicate that end-to-end latency was not determined by input-token count alone under the tested API conditions. For the four configurations subsequently evaluated dynamically, all measured mean response times were substantially shorter than the one-hour control interval used in the simulation. Nevertheless, these per-query values are specific to the tested model, API, prompts, hardware, and communication conditions and do not establish scalability or deployment reliability.

3.4. Qualitative analysis

To interpret the factors associated with the observed quantitative patterns, the reasoning responses of each configuration were examined in relation to the performance metrics. The Baseline configuration (B1), which lacks explicit reasoning path guidance, was excluded from the structural reasoning comparison and used primarily as a numerical reference. This section examines the reasoning characteristics and structural limitations of each configuration. Representative response samples are provided in Table A.2 to illustrate recurring reasoning patterns identified during qualitative inspection of the outputs.

3.4.1. Reasoning-category configurations

Zero-shot CoT (R1) demonstrated improved performance over B1 in reflecting occupant characteristics through a three-step reasoning path. Response analysis revealed a logical chain linking overweight BMI to increased insulation and reduced heat dissipation, supporting a downward setpoint adjustment. The reasoning also explicitly weighed the warm preference of older female occupants against the cooling preference associated with overweight BMI. However, adjustment values were expressed qualitatively without explicit numerical calculation, which may account for R1's higher MAE (0.67 °C) compared with R3.

Tree of Thought (R2) implemented a structure integrating the independent reasoning of three experts; however, response analysis revealed a structural limitation at the integration stage. Although each

Table 4
Response time, token usage, and estimated API cost by prompting configuration.

Category	ID	Configuration	Mean Response Time (s/query)	Input Tokens	Output Tokens	Estimated API Cost (USD/query)
Baseline	B1	Instruction-only	7.05	90	367	0.0039
Reasoning	R1	Zero-shot CoT	8.59	124	452	0.0048
	R2	Tree of Thought (ToT)	9.50	135	452	0.0049
	R3	Decomposition	11.32	156	549	0.0059
Context/ Knowledge	C1	RAG-style prompting	8.99	266	455	0.0052
	C2	ReAct-style prompting	11.26	142	498	0.0053
	C3	MRKL-style modular prompting	5.23	154	241	0.0028
Hybrid	R3+C3	Decomposition + MRKL-style modular	5.12	295	335	0.0041

expert independently derived setpoints based on physiological characteristics, environmental physics, and an integrative perspective, the final stage tended to converge toward an arithmetic compromise rather than a deeper comfort-bound evaluation. This suggests that in convergent tasks such as thermal setpoint generation, a divergent reasoning structure may induce inter-path interference and undermine the consistency of final outputs.

Decomposition (R3) exhibited the most explicitly structured reasoning workflow among the evaluated configurations. The [Occupant Analysis] stage derived a single offset from age, gender, and BMI; the [Environmental Analysis] stage estimated adjustments associated with the indoor environmental variables; and the [Synthesis] stage integrated these components using Ta as the reference point. This stepwise structure may have contributed to R3's low numerical error (MAE = 0.43 °C). However, because the configuration used the current air temperature rather than a PMV-neutral anchor as its reference point, its generated setpoints showed relatively low PCCR (72.4 %) when the current condition was not aligned with PMV neutrality.

3.4.2. Context/Knowledge-category configurations

RAG-style prompting (C1) demonstrated structured reasoning by leveraging pre-extracted knowledge chunks from ASHRAE 55–2020 [40] and ISO 8996:2021 [41], estimating comfort ranges from CLO/-MET and weighing conflicting occupant characteristics (e.g., female/elderly occupants preferring warmer conditions versus overweight occupants preferring cooler conditions). However, a passive tendency consistently emerged in which the model restricted active adjustment because current conditions already fell within the calculated comfort range, converging toward a value within that range rather than actively deriving an occupant-specific setpoint. This limitation may explain why C1′s PCCR (76.0 %) is lower than those of C2 and C3.

ReAct-style prompting (C2) implemented the most systematic verification process through an iterative Thought-Action-Observation-Decision loop. However, response analysis revealed that the Action stage tended to reference standards without introducing numerical criteria beyond what was already in the prompt, reflecting C2′s design intent of evaluating the reasoning loop structure itself without directly injected knowledge chunks, unlike C1. Nevertheless, C2 achieved a high PCCR (89.0 %), indicating that iterative verification can function as a conservative mechanism that suppresses extreme setpoints.

To reduce reliance on direct nonlinear numerical computation by the LLM, C3 uses a modular structure in which a pre-calculated PMV-neutral temperature serves as a comfort-bound anchor and the LLM performs occupant-trait adjustment around it. C3 recorded the highest observed PCCR (91.6 %). The controlled counterfactual analysis also showed that C3 produced the most consistent age- and gender-conditioned changes, together with directionally coherent median BMI shifts. However, qualitative inspection indicated that the [Adjustment_Module] primarily represented the directional tendencies associated with age, gender, and BMI and then integrated them into a single offset with limited numerical justification. This qualitative offset formulation may have contributed to the lower numerical agreement observed for C3 (MAE = 1.52 °C).

3.4.3. Trade-off and reasoning interaction in the hybrid configuration

The qualitative analysis revealed a clear distinction among the eight evaluated configurations between numerical agreement and PMV-based comfort-bound compliance (Table 5). R3, which recorded the lowest observed numerical error, showed lower PCCR because it used the current air temperature as the reference point without a PMV-neutral anchor. In contrast, C3, which recorded the highest observed PCCR, exhibited lower numerical agreement due to ambiguity in the occupant-trait offset calculation. Based on these complementary performance profiles, the hybrid configuration (R3+C3) was constructed after the initial seven-configuration comparison. In the static evaluation, the hybrid did not retain R3's lowest numerical error or C3's highest PCCR. Qualitative inspection suggested that combining current-temperature-based fine-tuning with PMV-neutral anchoring shortened parts of the decomposition process, as the model often treated MET and CLO as already incorporated into the anchor. This interaction may have limited the expected synergy of the hybrid structure, indicating that internal logical coherence and reference-point consistency can influence configuration performance. The static characteristics of B1, R3, C3, and R3+C3 are further examined through dynamic evaluation in Section 4.

4. Dynamic performance evaluation

4.1. Simulation environment

To evaluate the dynamic control performance of the four selected prompting configurations, the DOE Commercial Prototype Building Model for a Medium Office was selected as the target building, and EnergyPlus simulations were performed [47,48]. This model represents a typical three-story commercial office building equipped with a variable air volume (VAV) system with electric reheat, and the building's physical envelope specifications and default occupancy schedules were maintained at the DOE default settings to ensure a consistent simulation basis.

To support a balanced evaluation of control performance under both cooling and heating conditions based on the selected building model, the monthly distributions of Heating Degree Days (HDD) and Cooling Degree Days (CDD) by climate zone were analyzed in accordance with the climate zone classification defined in ASHRAE 169–2021 [49]. Specifically, the region with the most balanced annual distribution between heating-dominant months (HDD > 0 and CDD = 0) and

Table 5
Qualitative analysis summary of reasoning characteristics by prompting configuration.

ID	MAE / PCCR	Reasoning Strength	Structural Limitation	Interpretation of Performance Pattern
B1	1.35 °C / 81.9 %	None (reference configuration)	Weak and inconsistent occupant-trait responses	No explicit reasoning structure; inconsistent metadata use observed
R1	0.67 °C / 85.7 %	Occupant traits reflected via 3-step sequential reasoning	Adjustment values remain non-numerical; relies on qualitative expressions	Directional reasoning secured, but calculations remain opaque
R2	1.03 °C / 86.0 %	Multi-perspective path generation	Expert C reduces to a simple compromise; range-then-fix pattern	Divergent structure associated with inter-path compromise in a convergent task
R3	0.43 °C / 72.4 %	Per-variable numeric offsets and explicit formula integration	Uses current air temperature as reference; PMV compliance not guaranteed	High reference tracking, but no PMV-neutral anchor
C1	0.62 °C / 76.0 %	ASHRAE/ISO knowledge chunks and comfort-range reasoning	Passive convergence toward current conditions within comfort zone	Domain knowledge utilized, but limited active adjustment
C2	0.68 °C / 89.0 %	Iterative verification loop suppresses extreme setpoints	Action step references standards without additional numeric content	Conservative verification associated with high comfort-bound compliance
C3	1.52 °C / 91.6 %	PMV-neutral anchor associated with the highest observed PCCR	Offset remains qualitative and tends to converge toward the anchor	Strong PMV-based anchor, but limited occupant-trait quantification
R3+C3	1.50 °C / 88.1 %	PMV-neutral anchor + decomposition-based reasoning	Anchor reduces part of per-variable decomposition	Reference-point interaction limits simultaneous improvement

cooling-dominant months (CDD > 0 and HDD = 0) was identified. As a result, Atlanta, USA—corresponding to International Energy Conservation Code Climate Zone 3A—was selected as the final simulation location. Based on these quantitative criteria, a total of six months comprising a clearly delineated heating season (Jan, Feb, and Dec) and cooling season (Jun, Jul, and Aug) were designated as the primary analysis period, and Typical Meteorological Year 3 weather data [50] were applied to support reliable simulations.

The occupant profile used in the simulation was configured based on the statistical characteristics of the 1080-sample evaluation dataset described in Section 3, to maintain consistency with the LLM input variables. Occupant age was set to 24, while gender and BMI were fixed at the modal values of the dataset, corresponding to female and a Normal BMI range ($18.5 \leq \text{BMI} \leq 24.9$), respectively. MET was set to 1.1 met, representing typical sedentary office activity. CLO was differentiated by season to reflect seasonal clothing patterns, with values of 1.0 clo for the heating season and 0.5 clo for the cooling season, ensuring that the PMV calculations within EnergyPlus and the LLM reasoning shared identical occupant parameters. To reflect the thermal characteristics of different building orientations, the indoor space was divided into Perimeter Zones (12 zones) exposed to solar radiation and Core Zones (3 zones) dominated by internal heat gains. The mean values of T_a, T_r, and RH were then extracted and used as LLM inputs. Va was uniformly set to 0.2 m/s, consistent with the default specification of the DOE prototype building.

An integrated interface was constructed using the Callback functionality of the EnergyPlus Python API to implement closed-loop LLM-based dynamic control (Fig. 2). The control logic executes at the callback_begin_zone_timestep_before_set_current_weather point at every simulation timestep, and the LLM is called once per hour to account for API communication overhead and building thermal inertia. At each hourly control update, the extracted indoor environmental data and occupant characteristic variables are combined into a prompt and transmitted to the OpenAI API. The setpoint returned by the LLM is then applied to the system via the DualSetpoint Thermostat actuator. LLM control is deactivated during the Warmup Period, which is automatically executed by EnergyPlus prior to the start of the simulation for building thermal equilibrium convergence. The allowable setpoint range was constrained between 15.6 °C and 29.4 °C as a broad operational safety envelope to prevent unrealistic control outputs while preserving the setpoint-generation characteristics of each configuration, and setpoints resolved to one decimal place were used to compare the temperature generation tendencies among prompting configurations.

4.2. Control scenarios

The dynamic evaluation includes the original DOE prototype thermostat schedule as a reference case, together with four LLM-based configurations selected to represent each major category from the static evaluation: B1 (Baseline, absence of an explicit reasoning structure), R3 (Reasoning category, lowest observed static numerical error), C3 (Context/Knowledge category, highest observed static PCCR), and R3+C3 (Hybrid, integrating decomposition-based reasoning with modular anchoring). The hybrid configuration was added to examine whether the complementary static characteristics of R3 and C3 are preserved under closed-loop dynamic operation. The DOE prototype thermostat schedule was retained as a schedule-based contextual reference to the original prototype-building operation.

In addition, three occupant profiles with contrasting thermal-preference directions were established as combined-profile stress tests of closed-loop responsiveness (Table 6). Unlike the controlled counterfactual in Section 3.3.1, age, gender, and BMI varied simultaneously across these profiles. Case 1 represents the baseline profile, directly adopting the modal values from the evaluation dataset described in Section 3.1 (Age 24, Female, Normal BMI). Case 2 was designed as a young male/obese combination representing a relatively cool thermal preference, while Case 3 was designed as an elderly female/underweight combination representing a relatively warm thermal preference [11,12,41,51]. Case 2 and Case 3 were designed as contrastive stress-test profiles based on reported directional tendencies in age-, gender-, and

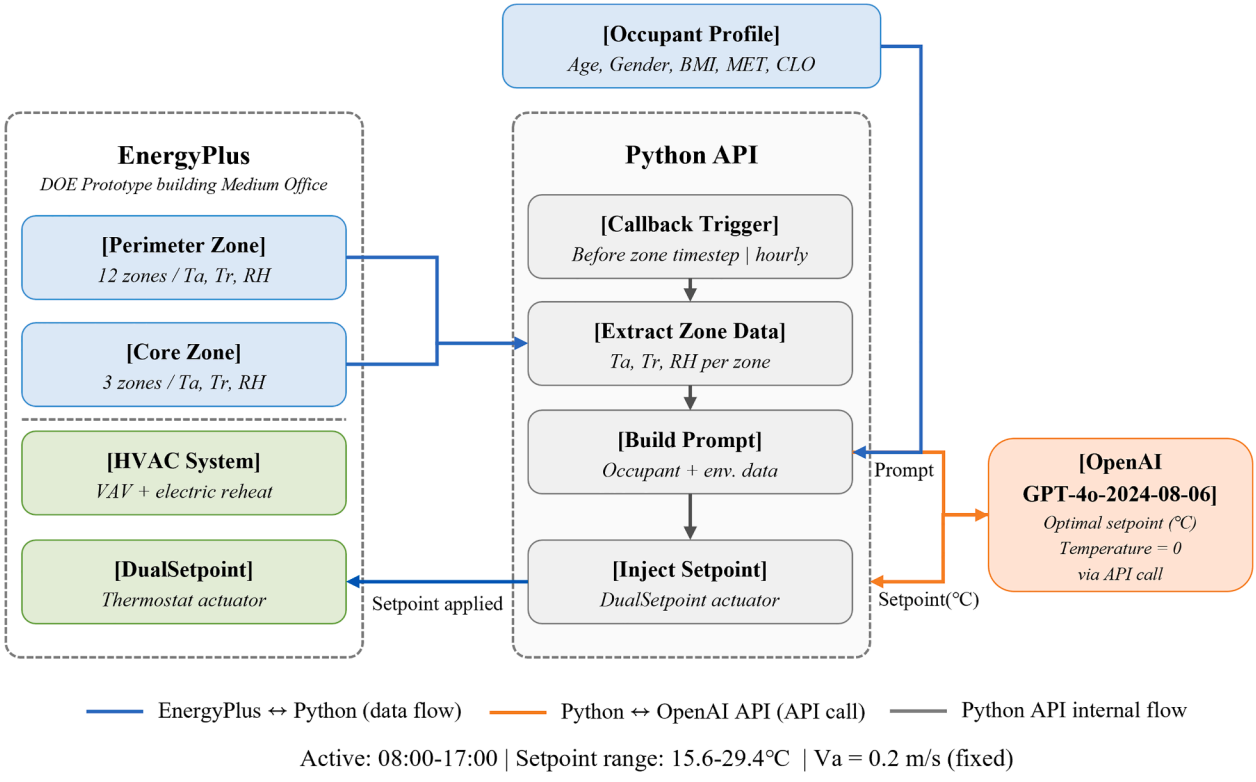


Fig. 2. System architecture of the LLM–EnergyPlus dynamic control interface.

Table 6
Occupant profiles for dynamic control scenarios.

	Case 1	Case 2	Case 3
Thermal Preference	Baseline	Cool Preference	Warm Preference
Age (years)	24	20	80
Gender	Female	Male	Female
BMI	Normal	Obesity	Underweight
MET (Heating, Cooling)	1.1 met	1.1 met	1.1 met
CLO (Heating)	1.0 clo	1.0 clo	1.0 clo
CLO (Cooling)	0.5 clo	0.5 clo	0.5 clo

BMI-related thermal perception, rather than as representative population groups. By aligning all variables within each profile (age, gender, and BMI) in the same thermal preference direction, the scenarios were designed to examine configuration-specific occupant responsiveness. The age and BMI of each profile were selected based on distributional statistics of the Section 3.1 evaluation dataset, specifically the modal and extreme values.

In the selected EnergyPlus model, occupant loads were represented using a fixed people-per-floor-area schedule, while age, gender, and BMI were not represented in the physical occupant model. Accordingly, the building and occupant-load settings were held constant across the three cases, and only the occupant-characteristic variables supplied to the LLM prompt were varied. LLM-based control was activated only during working hours (08:00–17:00, Monday through Friday) in the selected heating and cooling seasons; during inactive hours and weekends, the control logic reverted to the default thermostat schedule. Furthermore, the LLM-generated setpoint is applied uniformly to both the heating and cooling setpoints without a separate deadband in order to isolate the setpoint-generation characteristics of the prompting configurations.

The control performance of each scenario was evaluated from two perspectives: thermal comfort and energy use. Thermal comfort was quantified using CSR, defined as the proportion of time during which the indoor PMV satisfies the ±0.5 comfort range. Energy performance was assessed by measuring the total HVAC electricity consumption (kWh) of the building.

4.3. Results

The CSR and energy consumption values reported in the following analysis were calculated based on working hours (08:00–17:00) during the heating and cooling seasons when LLM control is activated. Thermal comfort was evaluated using the Fanger PMV model implemented in EnergyPlus, where age, gender, and BMI enter the dynamic evaluation through their influence on the LLM-generated setpoint. Since MET was held constant across all three occupant profiles and CLO was seasonally adjusted but identical across cases, differences in PMV arise from changes in the LLM-generated setpoint. CSR was calculated based on the hourly mean PMV of both Perimeter Zone and Core Zone. This approach was adopted because the LLM inputs represent the mean environmental data of these two zone groups, reflecting the thermal comfort condition of occupants across the entire building.

Table 7
Dynamic control results for Case 1 (Age 24, Female, Normal BMI).

Configuration	Season	Mean Setpoint (SD)	Mean CSR (%)	HVAC Electricity (kWh)
B1	Heating	23.19 (0.91)	90.7	4839.3
	Cooling	23.72 (0.67)	73.8	6282.4
R3	Heating	20.79 (1.44)	39.0	2558.8
	Cooling	26.34 (1.49)	72.5	5227.0
C3	Heating	25.11 (1.28)	94.0	8237.7
	Cooling	26.03 (0.59)	99.1	5095.0
R3+C3	Heating	25.15 (1.30)	94.0	8309.0
	Cooling	26.11 (0.49)	99.3	5040.9

4.3.1. Single occupant profile

Table 7 presents the dynamic control results for the heating and cooling seasons under the dataset-representative occupant profile (Case 1: Age 24, Female, Normal BMI). The results indicate that the static performance ordering of the four selected configurations was not preserved under closed-loop operation, suggesting that static numerical agreement alone did not reliably anticipate their dynamic PMV-based comfort satisfaction.

The most pronounced performance reversal in the heating season was observed for Decomposition (R3). Although R3 recorded the highest numerical agreement in static evaluation (MAE = 0.43 °C), its mean CSR in the dynamic heating environment was 39.0 %, the lowest among the four evaluated configurations. The setpoint standard deviation (SD) of R3 (1.44 °C) was higher than those of B1 (0.91 °C), C3 (1.28 °C), and R3+C3 (1.30 °C), and Fig. 3 shows that R3′s heating-season PMV distribution shifted toward the lower comfort boundary with a wider spread. R3 recalculates per-variable offsets at each hourly timestep using current environmental conditions as the reference point. The resulting outputs showed greater setpoint variability, which may have contributed to lower comfort satisfaction despite high static numerical agreement.

In contrast, Baseline (B1), which recorded the highest static numerical error (MAE = 1.35 °C), achieved a mean CSR of 90.7 % under dynamic heating conditions. Its comparatively low setpoint variability (SD = 0.91 °C) coincided with the high heating-season CSR observed in Case 1. These findings indicate a limited correspondence between high static numerical agreement with a reference dataset and improved dynamic comfort performance, supporting the independent importance of dynamic evaluation.

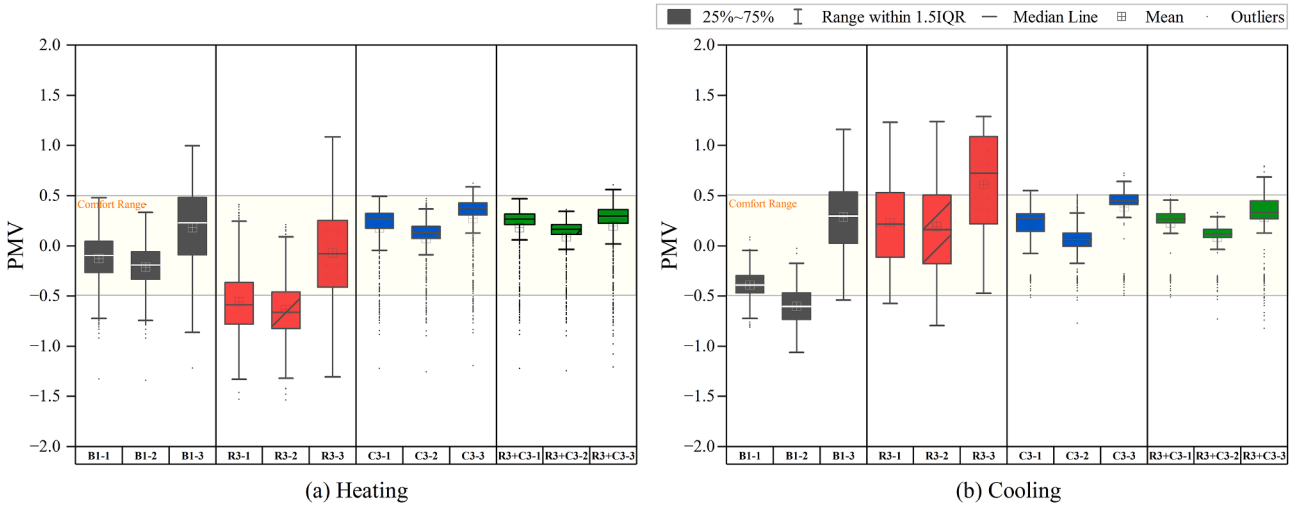
MRKL-style modular prompting (C3) and the hybrid R3+C3 achieved the highest heating-season mean CSR among the four evaluated configurations, both reaching 94.0 %. As shown in Fig. 3, C3 exhibited a narrower PMV distribution around the comfort range than R3. The higher CSR of C3 was accompanied by setpoints centered more closely around PMV neutrality, consistent with its use of a PMV-neutral anchor. This anchoring structure may have reduced violations of the defined comfort range under the tested dynamic conditions. Both PMV-anchored configurations also used substantially more heating electricity than B1 and R3 (C3: 8237.7 kWh; R3+C3: 8309.0 kWh). Thus, in Case 1 heating, higher CSR coincided with greater HVAC electricity use.

During the cooling season, the differences in CSR among the four evaluated configurations were also pronounced, with C3 and R3+C3 maintaining strong comfort performance while also showing lower HVAC energy use than B1. Specifically, C3 attained a mean CSR of 99.1 % and R3+C3 attained 99.3 %, compared to 73.8 % for B1 and 72.5 % for R3. In terms of HVAC energy consumption, the order was R3+C3 (5040.9 kWh), C3 (5095.0 kWh), R3 (5227.0 kWh), and B1 (6282.4 kWh).

4.3.2. Occupant responsiveness across diverse profiles

Table 8 presents a comparison of the dynamic control results across the three occupant profiles (Cases 1–3) for the four evaluated configurations. Fig. 4 illustrates the corresponding PMV-based comfort satisfaction rates across the occupant profiles for the heating and cooling seasons. During the heating season, R3 exhibited the largest variation in setpoints across the three occupant profiles (Case 2: 20.30 °C to Case 3: 23.67 °C, range = 3.37 °C), compared with C3 (range = 1.37 °C) and B1 (range = 2.30 °C). In terms of directionality, all four configurations lowered the setpoint for the cool-preference profile (Case 2) and raised it for the warm-preference profile (Case 3). Because age, gender, and BMI changed simultaneously and the indoor environment evolved through closed-loop feedback, these differences represent combined-profile dynamic responses rather than the independent effects of individual occupant characteristics.

However, the relationship between response magnitude and control outcomes differed substantially across configurations. R3 exhibited a



Note: B1-1, R3-1, C3-1, R3+C3-1 denote Case 1; B1-2, R3-2, C3-2, R3+C3-2 denote Case 2; B1-3, R3-3, C3-3, R3+C3-3 denote Case 3. The shaded area indicates the thermal comfort range (PMV ±0.5).

Fig. 3. PMV distribution by prompting configuration and occupant profile.

Table 8
Dynamic control results across three occupant profiles.

Configuration	Case	Season	Mean Setpoint (SD)	Mean CSR (%)	HVAC Electricity (kWh)
B1	Case 1	Heating	23.19 (0.91)	90.7	4839.3
		Cooling	23.72 (0.67)	73.8	6282.4
	Case 2	Heating	22.69 (1.05)	89.5	4817.4
		Cooling	22.89 (0.76)	31.0	7221.7
	Case 3	Heating	24.99 (1.52)	71.3	7228.0
		Cooling	26.46 (1.25)	69.4	5043.9
R3	Case 1	Heating	20.79 (1.44)	39.0	2558.8
		Cooling	26.34 (1.49)	72.5	5227.0
	Case 2	Heating	20.30 (1.25)	28.8	2601.1
		Cooling	26.06 (1.85)	71.4	5528.5
	Case 3	Heating	23.67 (2.48)	67.9	6183.8
		Cooling	27.92 (1.50)	37.8	6389.1
C3	Case 1	Heating	25.11 (1.28)	94.0	8237.7
		Cooling	26.03 (0.59)	99.1	5095.0
	Case 2	Heating	24.43 (1.32)	94.4	7384.6
		Cooling	25.39 (0.84)	98.5	5255.4
	Case 3	Heating	25.80 (1.35)	86.7	9164.3
		Cooling	26.77 (0.54)	63.6	5263.7
R3+C3	Case 1	Heating	25.15 (1.30)	94.0	8309.0
		Cooling	26.11 (0.49)	99.3	5040.9
	Case 2	Heating	24.54 (1.19)	94.8	7490.8
		Cooling	25.55 (0.69)	99.6	4987.5
	Case 3	Heating	25.29 (1.85)	90.2	8714.2
		Cooling	26.08 (2.18)	81.9	5515.9

pronounced upward adjustment in Case 3 relative to Case 1; however, it recorded the lowest heating-season mean CSR among the four evaluated configurations, with values of 39.0 % (Case 1), 28.8 % (Case 2), and 67.9 % (Case 3). R3 also showed the largest setpoint variability in Case 3 (SD = 2.48 °C). This variability coincided with lower heating-season CSR and more frequent PMV values below the comfort range. During the cooling season, the Case 3 setpoint of R3 rose to 27.92 °C, an increase of 1.58 °C relative to Case 1 (26.34 °C). As illustrated in Fig. 3, the PMV distribution of R3 Case 3 substantially exceeded the upper comfort boundary (+0.5), causing the mean CSR to drop to 37.8 %. This outcome is consistent with the absence of a PMV-neutral anchor in the current-temperature-based reference structure analyzed in Section 3. By contrast, B1 maintained the same directional pattern across the three profiles during the heating season, but its mean CSR decreased from 90.7 % in Case 1 to 71.3 % in Case 3. In Case 2 cooling, its mean setpoint of 22.89 °C was associated with a mean CSR of 31.0 % and PMV values below the comfort range, indicating overcooling in this scenario.

C3 exhibited a relatively small setpoint variation range across cases (heating: 1.37 °C, cooling: 1.38 °C) and maintained comparatively high heating-season mean CSR values across the three cases (Case 1: 94.0 %, Case 2: 94.4 %, Case 3: 86.7 %). This stability may be attributed to the PMV-neutral anchor, which provides a comfort-bound reference point at each timestep, with occupant characteristic offsets applied as relatively small adjustments around this baseline. In the cooling season, C3 maintained high mean CSR in Case 1 (99.1 %) and Case 2 (98.5 %), but a notable decline was observed in Case 3 (63.6 %). The higher setpoint in Case 3 coincided with more hours above the PMV upper bound, suggesting that the combined warm-preference adjustment was too large for this tested condition. Because age, gender, and BMI varied simultaneously, this result does not isolate the contribution of any individual characteristic.

The hybrid configuration (R3+C3) maintained high mean CSR across all three cases in the heating season (94.0 %, 94.8 %, and 90.2 %) and in Cases 1 and 2 of the cooling season (99.3 % and 99.6 %), following a pattern similar to C3 under the baseline and cool-preference profiles. In Case 3 cooling, R3+C3 achieved a mean CSR of 81.9 %, compared with 63.6 % for C3. This higher CSR was accompanied by a lower mean setpoint for R3+C3 than for C3 (26.08 °C vs. 26.77 °C). These results suggest that, under this tested condition, the hybrid structure moderated the upward setpoint adjustment observed for C3.

4.3.3. *Energy–comfort trade-off*

While the static evaluation in Section 3 identified a distinction between numerical agreement (MAE, RMSE) and PMV-based comfort-bound compliance (PCCR), the dynamic simulation results demonstrate that this distinction extends further across occupant profiles, seasonal conditions, and feedback-loop behavior. Fig. 5 illustrates the

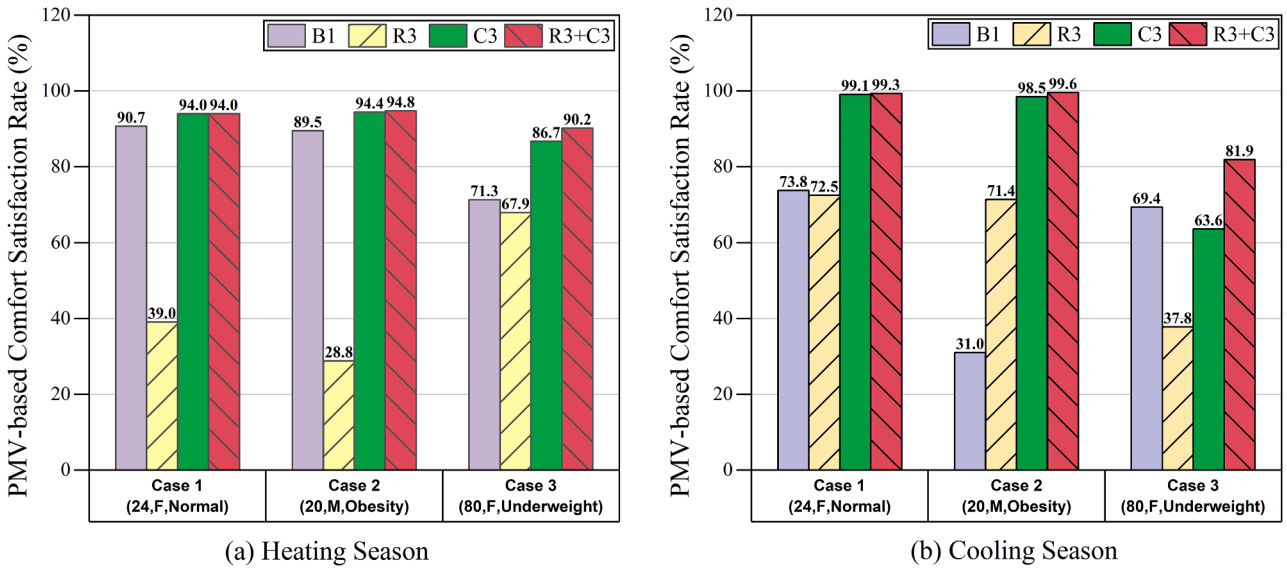
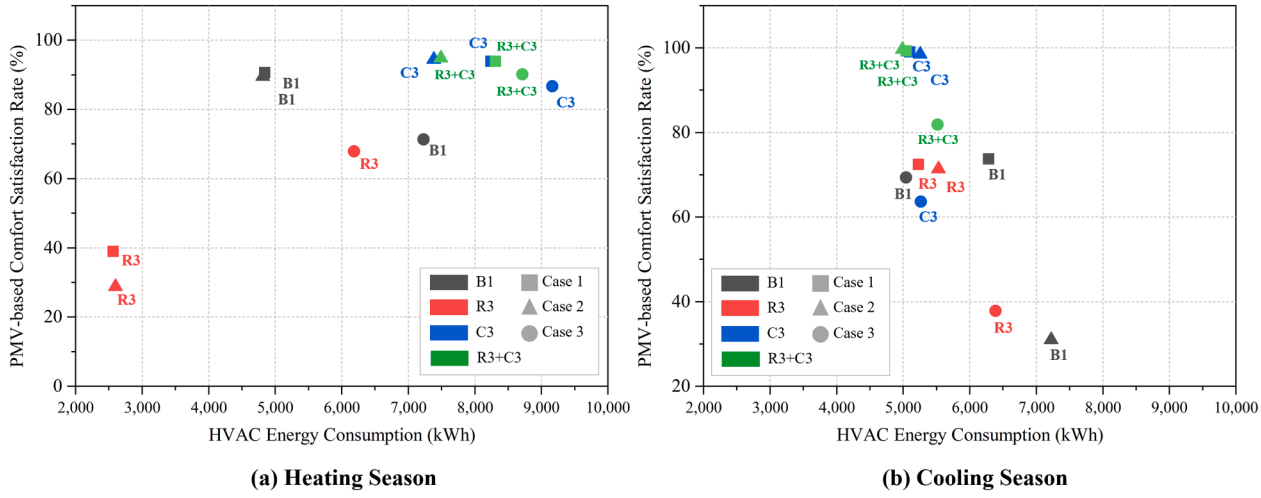


Fig. 4. PMV-based comfort satisfaction rate by prompting configuration and occupant profile.



Note: PMV-based comfort satisfaction rate corresponds to CSR as defined in Section 4.3.

Fig. 5. Representative energy-comfort trade-off by prompting configuration and occupant profile.

representative energy-comfort positioning of B1, R3, C3, and R3+C3 across the three profiles for both heating and cooling seasons, while Table 9 summarizes the aggregate dynamic results for the DOE prototype schedule reference and the selected LLM-based configurations.

Table 9
Aggregate dynamic outcomes for the DOE schedule reference and selected LLM-based configurations.

Configuration	Mean CSR (%)	Combined CSR (%)	Mean HVAC Electricity (kWh)	Mean Setpoint (°C)
DOE	84.2	78.7	4067.1	Schedule-based
B1	71.0	63.1	5905.4	23.99
R3	52.9	48.6	4748.0	24.18
C3	89.4	82.0	6733.5	25.59
R3+C3	93.3	89.2	6676.4	25.45

Note: Mean CSR refers to the arithmetic mean of the per-scenario CSR values across the six case–season combinations. Combined CSR refers to the proportion of timesteps in which both the Perimeter Zone and Core Zone PMVs simultaneously satisfied the ±0.5 comfort range, averaged across the six scenarios. The DOE schedule is reported as a schedule-based contextual reference rather than as an optimized conventional-control baseline.

Based on the aggregate results, the original DOE prototype thermostat schedule achieved a mean CSR of 84.2 % and the lowest HVAC electricity consumption (4067.1 kWh); these values are reported as a schedule-based contextual reference for the LLM-generated outcomes. Among the selected LLM configurations, R3 showed the lowest mean HVAC electricity use (4748.0 kWh) but also the lowest mean CSR (52.9 %), indicating limited correspondence between static reference-temperature tracking and dynamic comfort satisfaction. C3 achieved a mean CSR of 89.4 %, while R3+C3 achieved the highest mean CSR (93.3 %) and combined CSR (89.2 %) among the four tested configurations. C3 and R3+C3 also used more HVAC electricity than the DOE schedule and R3 (6733.5 and 6676.4 kWh, respectively), showing that higher PMV-based comfort satisfaction did not correspond to lower energy use under all tested conditions.

Taken together, the selected configurations occupied different positions across the reported evaluation dimensions. R3 showed strong reference-temperature tracking in static evaluation but greater setpoint

variability and lower comfort satisfaction in the tested heating cases. C3 showed high PMV-based comfort compliance but higher heating energy use and over-adjustment in the warm-preference cooling case. R3+C3 did not improve static numerical agreement over R3 but achieved the highest aggregate dynamic comfort satisfaction among the four tested configurations. Accordingly, these results characterize B1, R3, C3, and R3+C3 under the specified conditions rather than provide an exhaustive closed-loop ranking of all eight configurations.

5. Discussion

The present study compared all eight prompting configurations in static evaluation and examined four selected configurations in closed-loop dynamic simulation. While prior studies have identified LLMs as a promising approach for interpreting relationships between occupant comfort and physical environmental variables [52,53], systematic comparisons of prompting configurations within thermal environment control remain limited. The present study addresses this gap by demonstrating configuration-dependent differences in static outcomes across all eight configurations and in dynamic control behavior across the four selected configurations.

The key finding of the static evaluation is that numerical agreement (MAE and RMSE) and the PMV Comfort Compliance Rate (PCCR) were associated with different reference-point structures. Decomposition (R3), which decomposes and sums per-variable offsets using the current air temperature as the reference point, recorded the lowest observed MAE (0.43 °C). In contrast, MRKL-style modular prompting (C3), which uses a PMV-neutral anchor calculated from the Fanger PMV model, recorded the highest observed PCCR (91.6 %). These contrasting reference-point structures were consistent with the different performance profiles observed across the two evaluation dimensions.

The PMV-filter sensitivity analysis reinforced this distinction. In the 762 PMV-inconsistent TSV-neutral samples, R3 continued to track the observed neutral temperatures most closely, whereas C3 and R3+C3 remained closest to PMV neutrality according to PCCR, mean absolute PMV, and mean PPD. The persistence of high PCCR for the PMV-anchored configurations, despite their substantially larger errors relative to the observed TSV-neutral temperatures, indicates that the compliance ranking was influenced by the shared PMV basis. Accordingly, PCCR should be interpreted as PMV-bound compliance rather than as independent evidence of personalized or subjectively perceived comfort.

The hybrid configuration (R3+C3), which combines decomposition-based reasoning with PMV-neutral anchoring, exhibited a distinct interaction between the two reference-point structures. Qualitative inspection suggested that the LLM often treated MET and CLO as already represented in the PMV-neutral anchor, which was accompanied by a shortened decomposition process. The hybrid showed lower static numerical agreement than R3 and lower PCCR than C3. These findings suggest that internal logical coherence and reference-point consistency should be considered when integrating multiple prompting structures. These static trade-offs did not translate directly to the dynamic evaluation, in which R3+C3 achieved the highest aggregate comfort satisfaction among the four tested configurations.

The controlled counterfactual analysis showed that metadata responsiveness depended on prompt structure. C3 produced the clearest overall pattern, with fully consistent age- and gender-conditioned changes and directionally coherent median BMI shifts, whereas R3+C3 and R1 were comparatively stable mainly for age and BMI. C2 and R3 showed weak or inconsistent responses, while R2 produced BMI-conditioned changes that did not follow a coherent expected direction. Although the warmer age and cooler higher-BMI directions observed in C3 and R3+C3 were broadly aligned with group-level tendencies reported in previous thermal-comfort studies [11–14,51], C1 and R3+C3 explicitly contained directional guidance. These findings therefore indicate configuration-dependent metadata responsiveness rather than validated physiological personalization.

The dynamic simulation further elucidated temporal control behavior that was not revealed in the static evaluation. R3, which recorded the lowest static numerical error, showed greater temporal setpoint variability and lower mean CSR across the tested heating profiles. In contrast, B1, which recorded the highest static numerical error, achieved comparatively stable comfort performance in some heating cases. For the four selected configurations, these results show that static numerical agreement alone did not reliably anticipate closed-loop control behavior, supporting the complementary value of dynamic evaluation.

Among the four dynamically evaluated configurations, those using PMV-neutral anchoring showed higher aggregate PMV-based comfort satisfaction, although their energy implications varied by season and profile. In the warm-preference stress-test profile (Case 3), C3 showed lower cooling-season comfort satisfaction, indicating that offset ambiguity can interact with dynamic HVAC conditions. These findings indicate that configuration performance varied with occupant profile, season, and control priority under the tested conditions and support further evaluation of adaptive selection strategies in comparisons with established controllers and empirical settings. The approach adopted in this study provides a mechanism for conditioning setpoint generation on occupant characteristics that are not direct inputs to standardized comfort indices, while its practical relevance should be further examined through empirical validation.

Several limitations should be noted. First, the primary static evaluation used GPT-4o [44], while Claude Opus 4.5 [45] and Llama-3.3–70B-Versatile [46] were evaluated on a smaller matched subset. R3 retained the lowest observed numerical-error ranking, but the remaining rankings, particularly for PCCR, were partly model dependent. Each sample–configuration pair was generated once; therefore, the bootstrap intervals represent sampling variability conditional on the observed outputs rather than run-to-run LLM variability.

Second, the dynamic evaluation was limited to B1, R3, C3, and R3+C3 in one DOE Medium Office model, one climate, and three combined occupant profiles. The findings should not be generalized to R1, R2, C1, or C2 or to other buildings and climates. The EnergyPlus occupant model was held constant across profiles, so age, gender, and BMI affected only the LLM-generated setpoints. The counterfactual analysis isolated these metadata under 12 fixed environmental conditions but evaluated response consistency rather than physiological or perceptual validity.

Third, PCCR, mean absolute PMV, mean PPD, and CSR share the Fanger PMV model. The 762-sample sensitivity analysis showed that PMV-anchored configurations could retain high PMV compliance despite large errors relative to observed TSV-neutral temperatures. These metrics therefore characterize PMV-based compliance and do not independently validate personalized comfort.

Finally, the control implementation omitted deadbands, actuator-rate constraints, failure handling, latency management, and cybersecurity safeguards. The DOE schedule was included only as a contextual reference; matched comparisons with fixed-setpoint, rule-based, PMV-based supervisory, and model predictive controllers are required to assess the practical added value of LLM-based control. Other prompting categories also remain outside the present scope.

6. Conclusion

This study quantitatively compared eight LLM prompting configurations for incorporating occupant characteristics, including age, gender, and BMI, into occupant-centric thermal control. All eight configurations were evaluated statically using the ASHRAE Global Thermal Comfort Database II [54], and four selected configurations were examined dynamically using EnergyPlus. The principal conclusions are as follows.

First, numerical agreement and PMV-based comfort-bound

compliance were shown to represent distinct evaluation dimensions. R3, which decomposed the setpoint-generation process around the current air temperature, recorded the lowest observed static error, whereas C3, which used a PMV-neutral anchor, recorded the highest observed PCCR. The hybrid R3+C3 configuration showed a separate static performance profile, indicating that internal coherence of the reference-point structure is important in prompting design. Sensitivity analysis using the PMV-inconsistent TSV-neutral subset did not change the leading configuration under each metric, but it amplified the distinction between agreement with observed neutral temperatures and compliance with PMV-based comfort bounds. Controlled counterfactual testing further showed that age-, gender-, and BMI-conditioned setpoint responses were configuration-dependent, with C3 showing the clearest overall metadata responsiveness across the three characteristics.

Second, among the four dynamically evaluated configurations, the static performance ordering was not preserved under closed-loop operation. R3's high static numerical agreement did not translate into high PMV-based comfort satisfaction, whereas the PMV-anchored C3 and R3+C3 configurations showed stronger comfort-bound behavior. These findings are limited to B1, R3, C3, and R3+C3 and do not establish the dynamic performance of R1, R2, C1, or C2.

Third, among the four dynamically evaluated configurations, the two PMV-anchored configurations achieved higher aggregate PMV-based comfort satisfaction than B1 and R3. However, C3 and R3+C3 also used more HVAC electricity than R3 and the DOE schedule reference. These results characterize a comfort–energy trade-off under the tested conditions and do not establish superiority over conventional control strategies.

The present study provides a systematic basis for comparing LLM prompting configurations in occupant-centric thermal setpoint generation. Future research should extend matched testing across additional LLMs, dynamically evaluate the remaining configurations, quantify repeated-generation variability, compare LLM-based control with established controllers, and validate metadata-conditioned setpoints using personalized comfort models, subjective feedback, and real-building measurements.

CRediT authorship contribution statement

Ji Young Yun: Writing – original draft, Validation, Methodology, Formal analysis, Conceptualization. **Eun Ji Choi:** Writing – review & editing, Methodology, Investigation, Conceptualization. **Jin Woo Moon:** Supervision, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

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Data availability

Data will be made available on request.

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